

Panoramic Radiograph y



- ▶ **Panoramic radiography:** Panoramic imaging (also called pantomography) is a technique for producing a single tomographic image of the facial structures that includes both the maxillary and mandibular dental arches and their supporting structures.



Principles of panoramic radiography

- ▶ x-ray source rotates around the patient's head and in opposite direction to the rotation of image receptor and collimator.
- ▶ Lead collimators in the shape of a slit, located at the x-ray source and at the image receptor, **limit the central ray to a narrow vertical beam**. Another collimator between the objects and the receptor reduces scattered radiation.



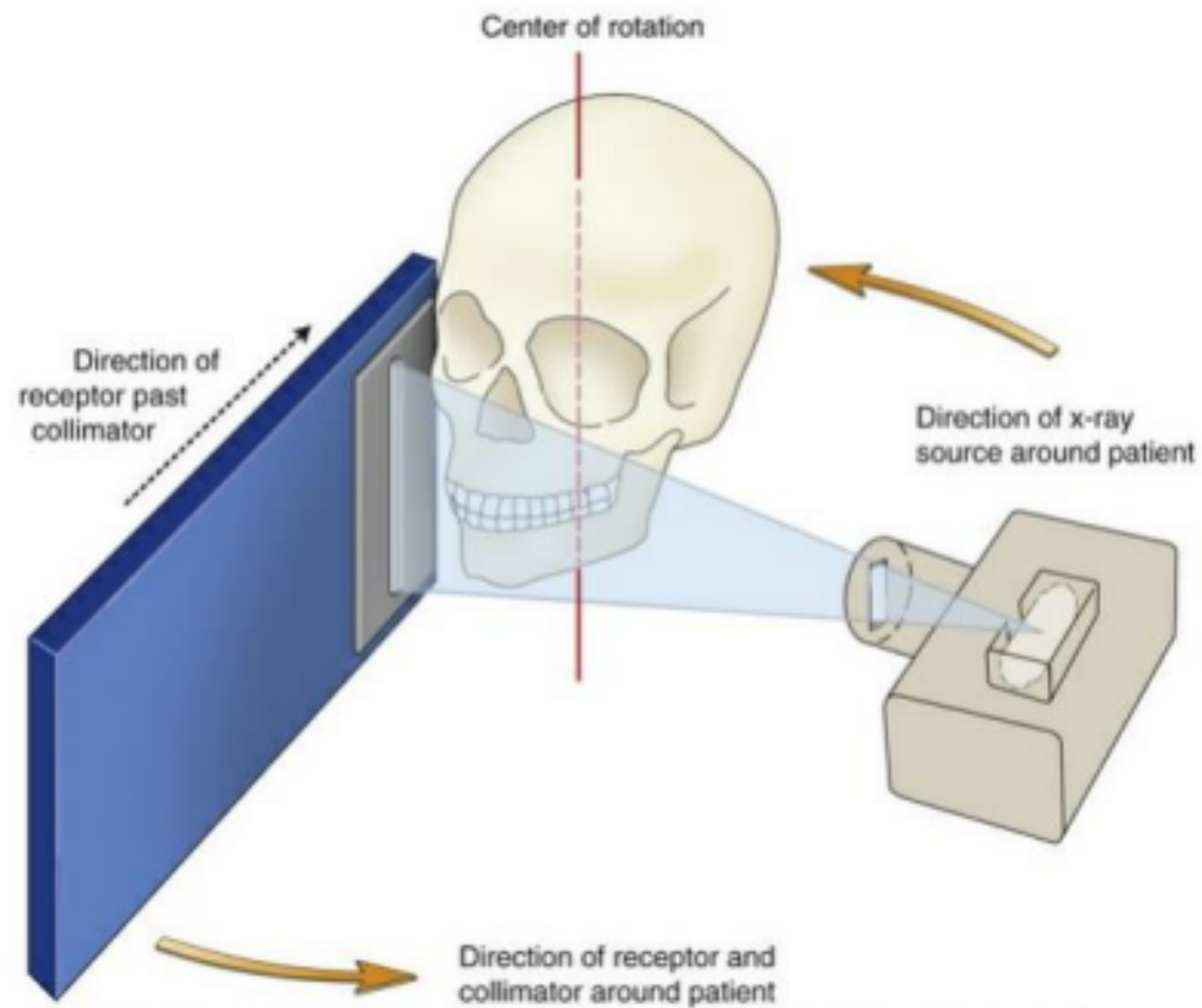


FIG. 8.8. Observation of relationship between the x-ray source, patient, and receptor.

Indications

- ▶ 1. Overall evaluation of dentition
- ▶ 2. Evaluation of intraosseous **pathology** such as cysts, tumors and infections.
- ▶ 3. Gross evaluation of **of tempromandiular joints**
- ▶ 4. Evaluation of **impacted teeth, dental implant, and orthodontic treatment**

- ▶ 5. Evaluation of permanent teeth **eruption and mixed dentition.**
- ▶ 6. Dentomaxillofacial **trauma** such as fracture.
- ▶ 7. Developmental **disturbances** of maxillofacial skeleton

Advantages of Panoramic radiograph

- ▶ 1. Broad coverage of facial bones and teeth
- ▶ 2. Low radiation dose
- ▶ 3. Ease of panoramic radiographic technique
- ▶ 4. Can be used in patients with **trismus** or in patients who cannot tolerate intraoral radiography
- ▶ 5. Quick and convenient radiographic technique
- ▶ 6. Useful visual aid in **patient education and case presentation**

Disdvantages

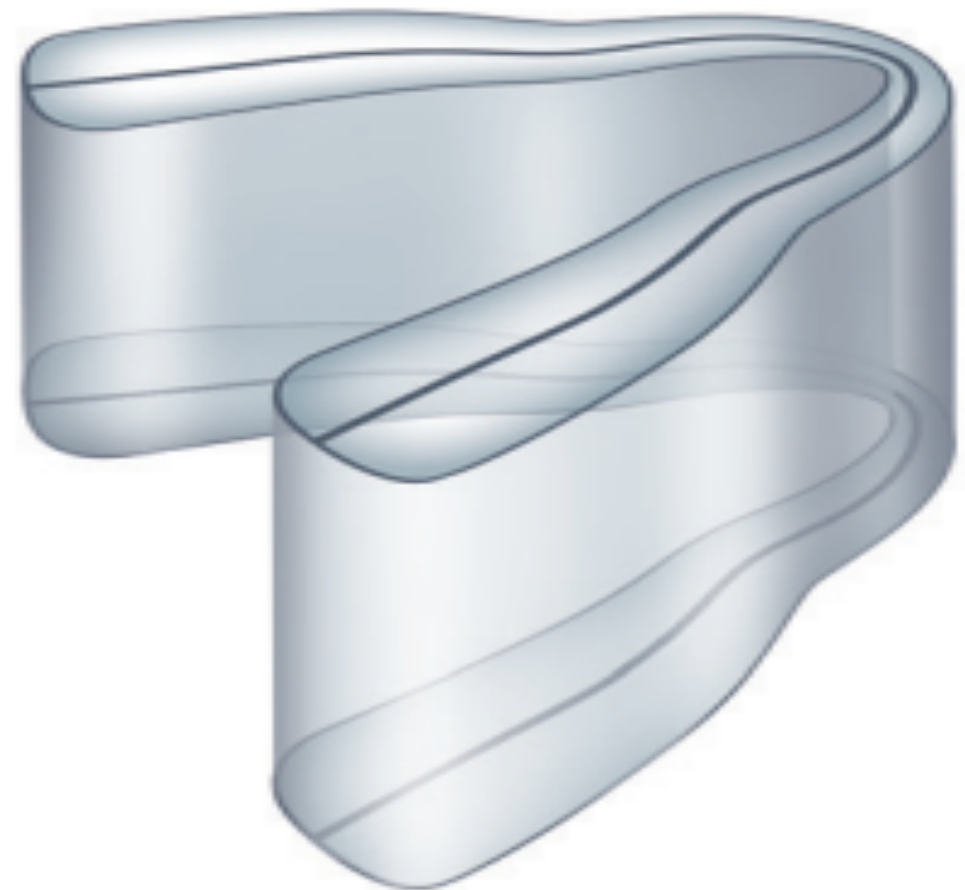
- ▶ 1. **Lower-resolution** images that do not provide the fine details provided by intraoral radiographs 2.
Magnification across image is unequal, making linear measurements unreliable
- ▶ 3. Image is **superimposition** of real, double, and ghost images and requires careful visualization to decipher anatomic and pathologic details



- ▶ 4. Requires accurate patient positioning to avoid positioning errors and artifacts, so it is **not suitable for children under 5 years or on some disabled patients.**
- ▶ 5. Difficult to image both jaws when patient has severe maxillomandibular discrepancy

Focal Trough (Image Layer)

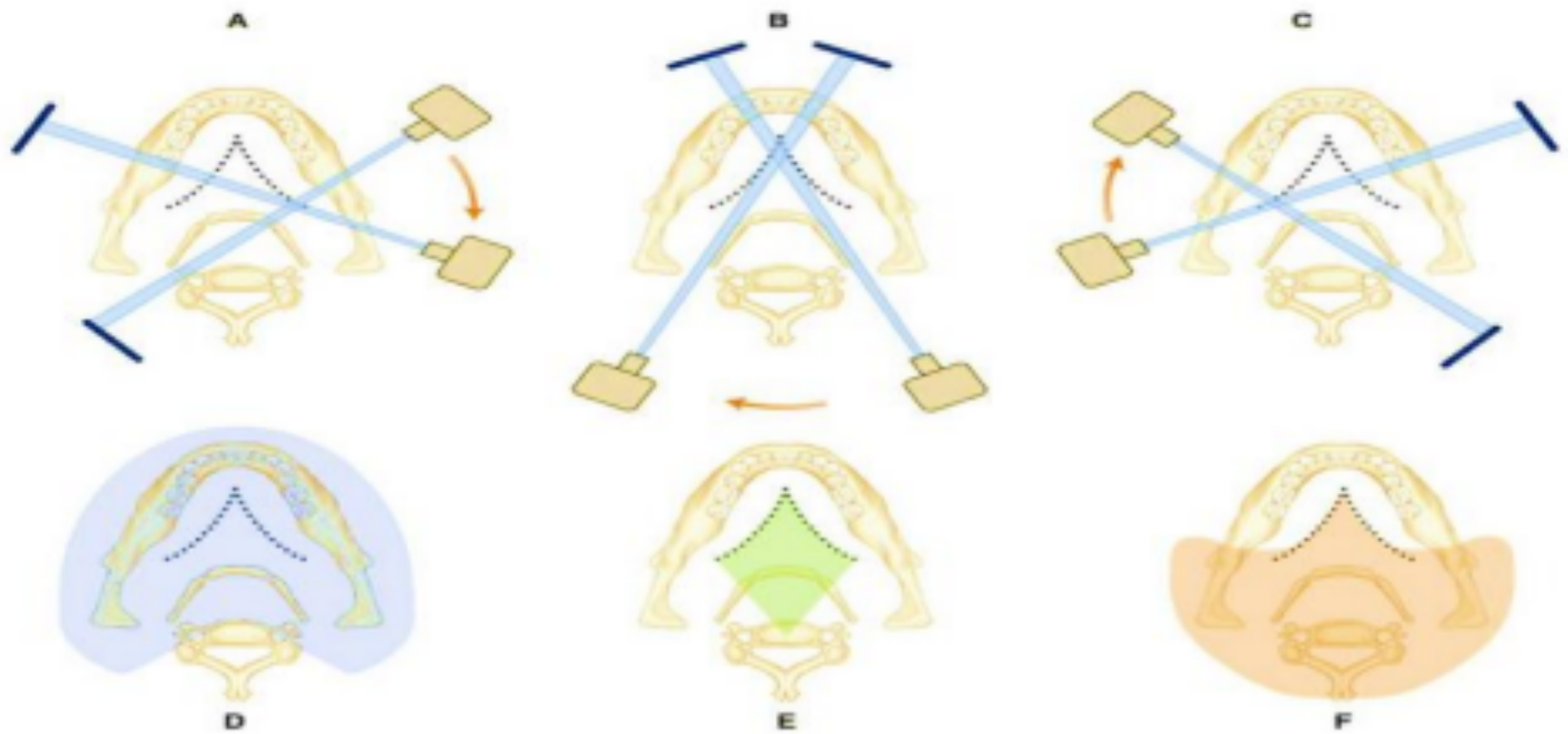
- ▶ is a wide, three-dimensional curved zone, where the structures lying within this layer are reasonably well defined on panoramic image.



• **Figure 12.3** Image layer, or focal trough.

Because of the rotational nature of x-ray source and receptor, the x-ray beam intercepts some anatomic structures twice.

- ▶ 1. **Real images**: object that lie between the center of rotation and the receptor
- ▶ 2. **Double images**: objects that lie posterior to the center of rotation
- ▶ 3. **Ghost images**: objects that located between the x-ray source and center of rotation, can cast ghost images. The ghost image appear on the opposite side of it's true anatomic location and at higher level.



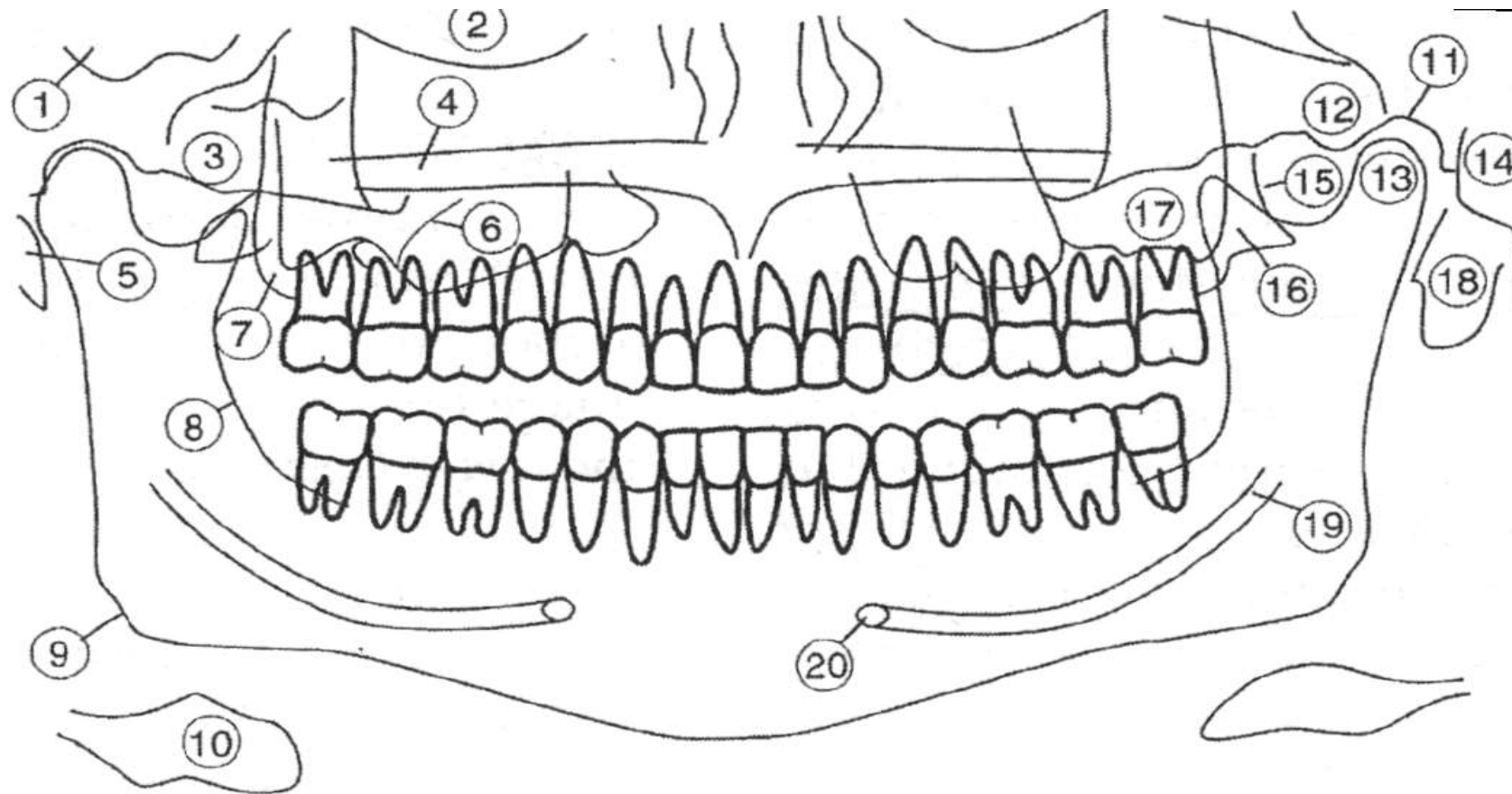
Technique, Patient Positioning and Head Alignment

- ▶ This technique utilises a narrow vertical negatively angled beam. The angle can be (-4 to -7°). The beam is shaped by a lead collimator which is long, narrow slit located at the X-ray source and the image receptor (film or digital plate).

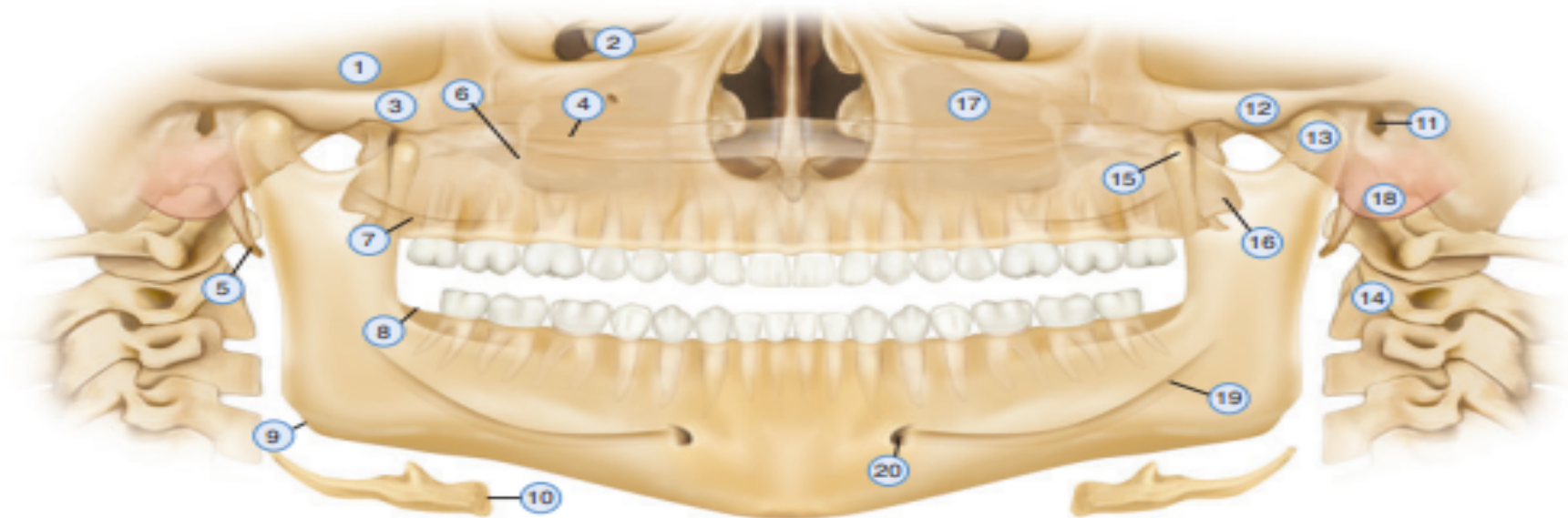
- ▶ 1. Dental appliances, earrings, and any other metallic objects in the head and neck region should be removed.
- ▶ 2. Demonstrate the machine to the patient
- ▶ 3. The anteroposterior position of the patient head is achieved typically by place the incisal edges of their maxillary and mandibular incisors into a notched positioning device (the biteblock).

- ▶ 4. The midsagittal plane must be centered within the focal trough without any lateral shift in the mandible
- ▶ 5. The occlusal plane is aligned so that it is lower anteriorly.
- ▶ 6. backs and spines as erect as possible and their necks extended.
- ▶ 7. Ask the patient to swallow and hold the tongue on the roof of the mouth.

ANATOMICAL LANDMARK



- | | | | |
|------------------------|----------------------------|-----------------------|---------------------|
| 1 Middle cranial fossa | 6 Septa in maxillary sinus | 11 Glenoid fossa | 16 Pterygoid plates |
| 2 Orbit | 7 Maxillary tuberosity | 12 Articular eminence | 17 Maxillary sinus |
| 3 Zygomatic arch | 8 External oblique line | 13 Mandibular condyle | 18 Ear lobe |
| 4 Palate | 9 Angle of mandible | 14 Vertebra | 19 Mandibular canal |
| 5 Styloid process | 10 Hyoid bone | 15 Coronoid process | 20 Mental foramen |

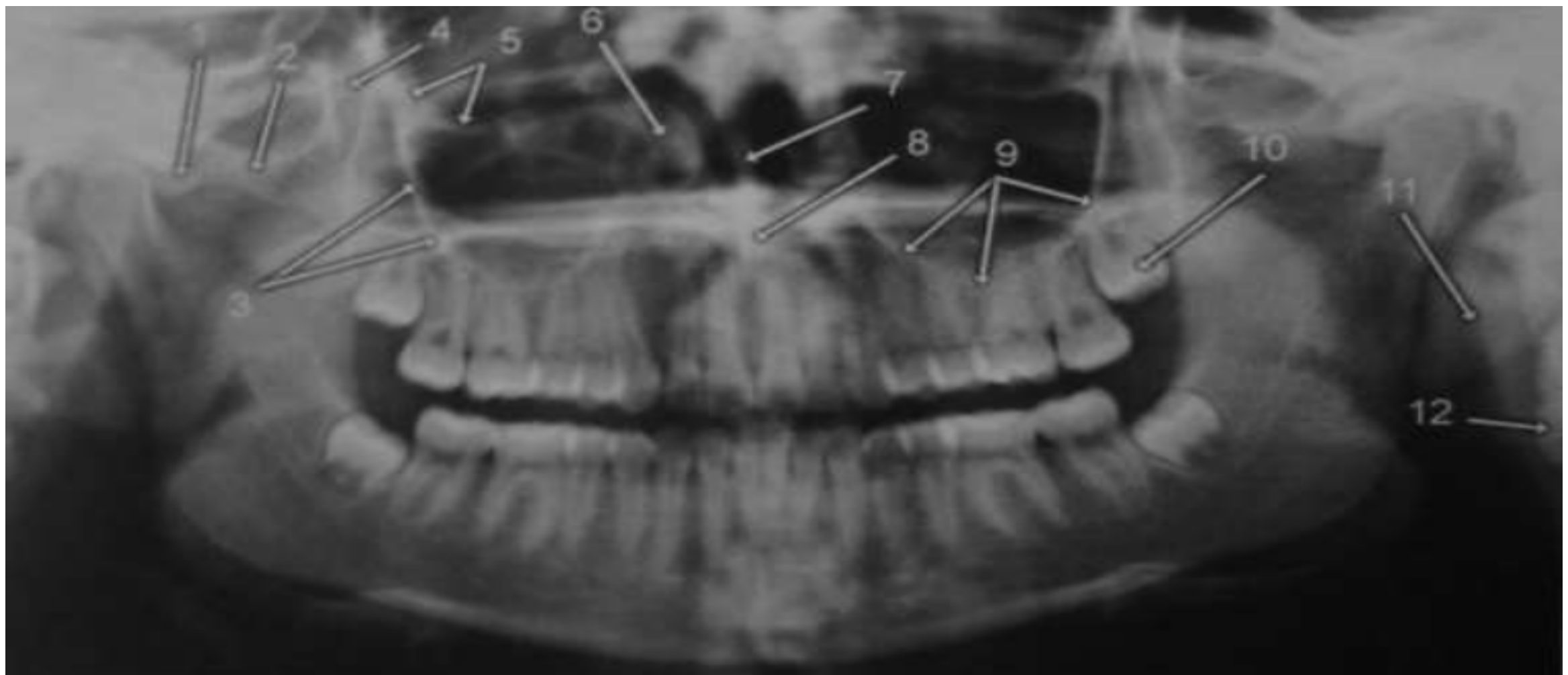


- | | |
|----------------------------|----------------------|
| ① Middle Cranial Fossa | ⑪ Glenoid Fossa |
| ② Orbit | ⑫ Articular Eminence |
| ③ Zygomatic Arch | ⑬ Mandibular Condyle |
| ④ Palate | ⑭ Vertebra |
| ⑤ Styloid Process | ⑮ Coronoid Process |
| ⑥ Septa in Maxillary Sinus | ⑯ Pterygoid Plates |
| ⑦ Maxillary Tuberosity | ⑰ Maxillary Sinus |
| ⑧ External Oblique Ridge | ⑱ Ear Lobe |
| ⑨ Angle of Mandible | ⑲ Mandibular Canal |
| ⑩ Hyoid Bone | ⑳ Mental Foramen |

Mandibular Structures

- 1 condyle
- 2 neck
- 3 sigmoid notch
- 4 coronoid process
- 5 ramus
- 6 inferior dental canal
- 7 angle
- 8 body
- 9 mental foramen
- 10 symphysis mentis
- 11 external oblique ridge



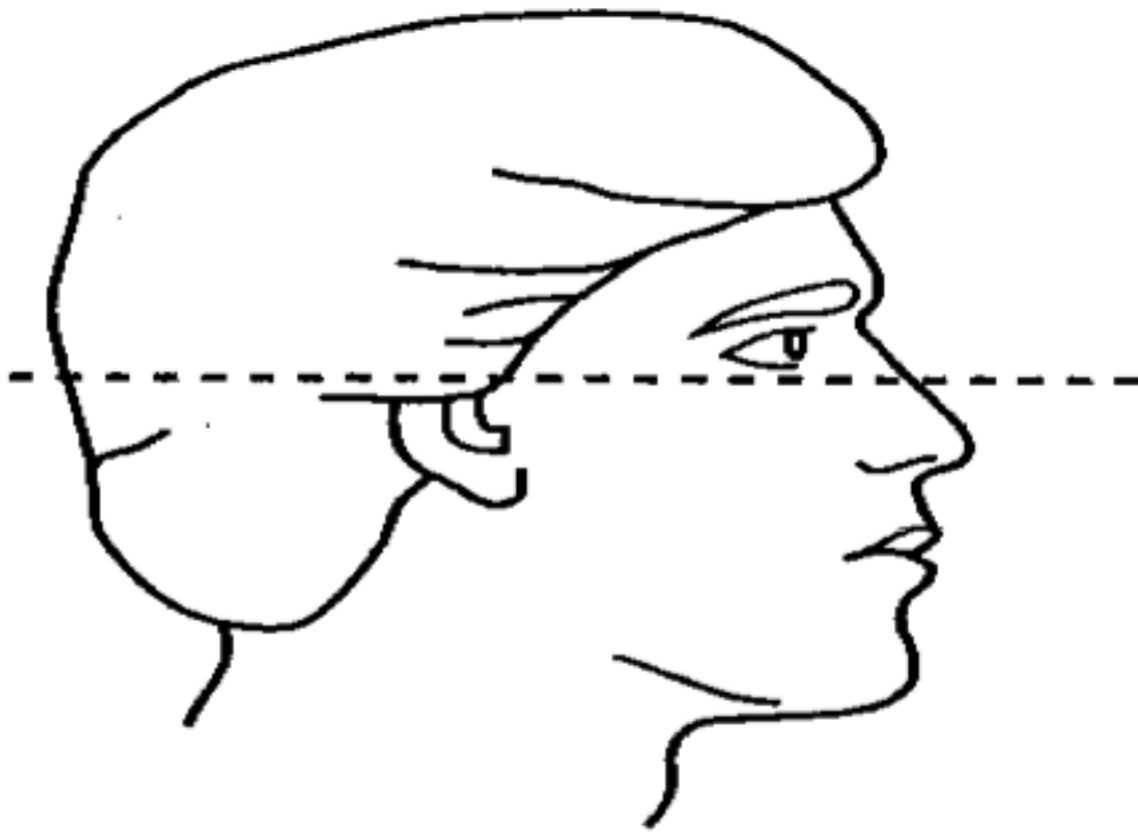


- | | |
|---------------------------------|-----------------------------|
| 1. articular tubercle | 8. anterior nasal spine |
| 2. zygomatic arch | 9. floor of maxillary sinus |
| 3. zygomatic process of maxilla | 10. developing third molar |
| 4. pterygomaxillary fissure | 11. ear lobe |
| 5. orbital rim | 12. cervical vertebra |
| 6. inferior nasal concha | |
| 7. nasal septum | |

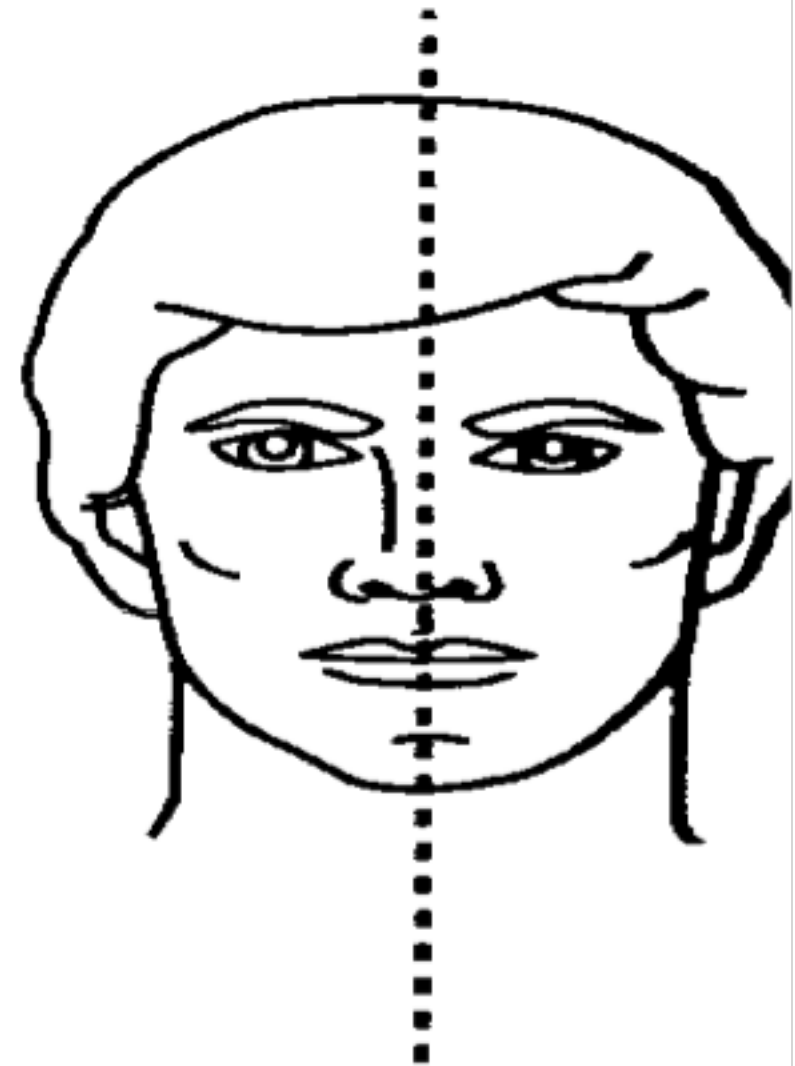
Equipment







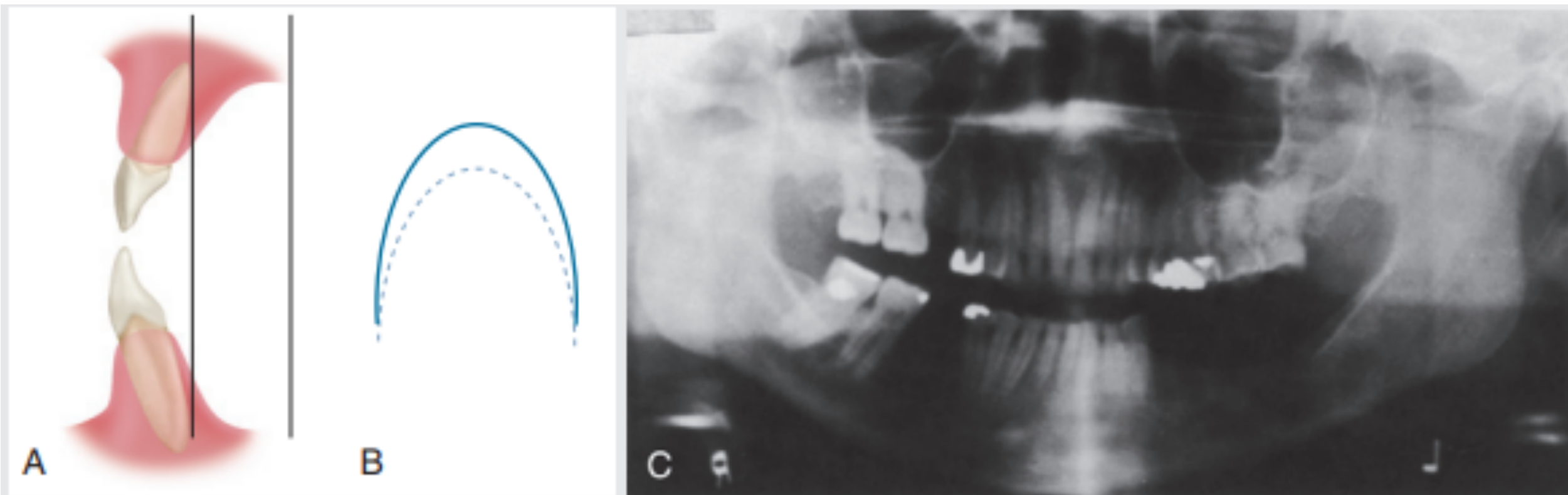
Frankfort plane



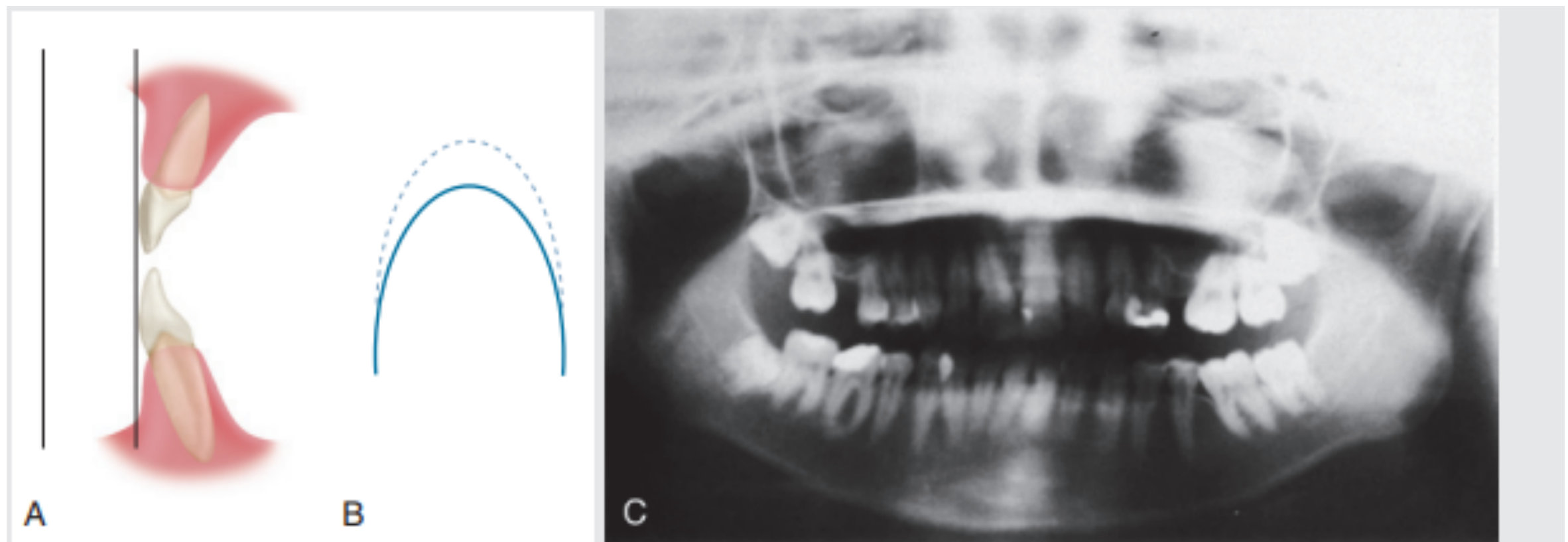
Midsagittal plane



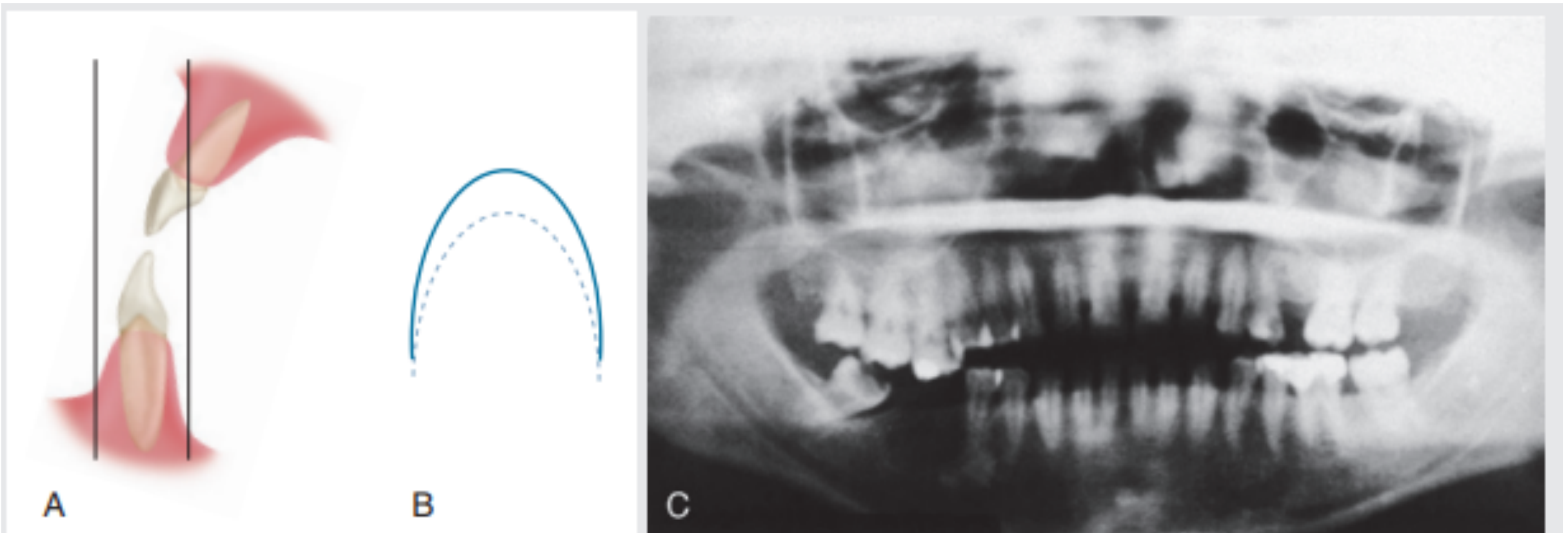
Errors



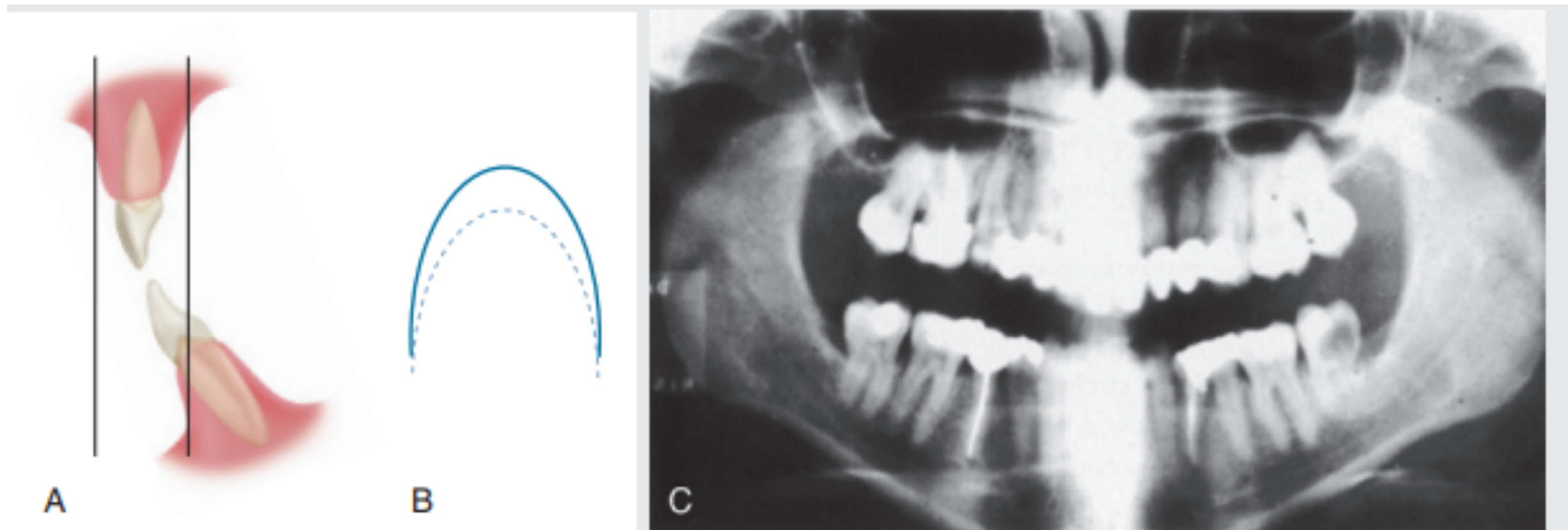
• **Figure 12.26** Patient positioned too far forward. Anterior teeth appear blurred and narrow. Spinal column is superimposed on the ramus, and premolars are overlapped. **A**, Anterior teeth forward of the focal plane. **B**, Dental arch (solid line) anterior to focal plane (broken line). **C**, Radiograph.



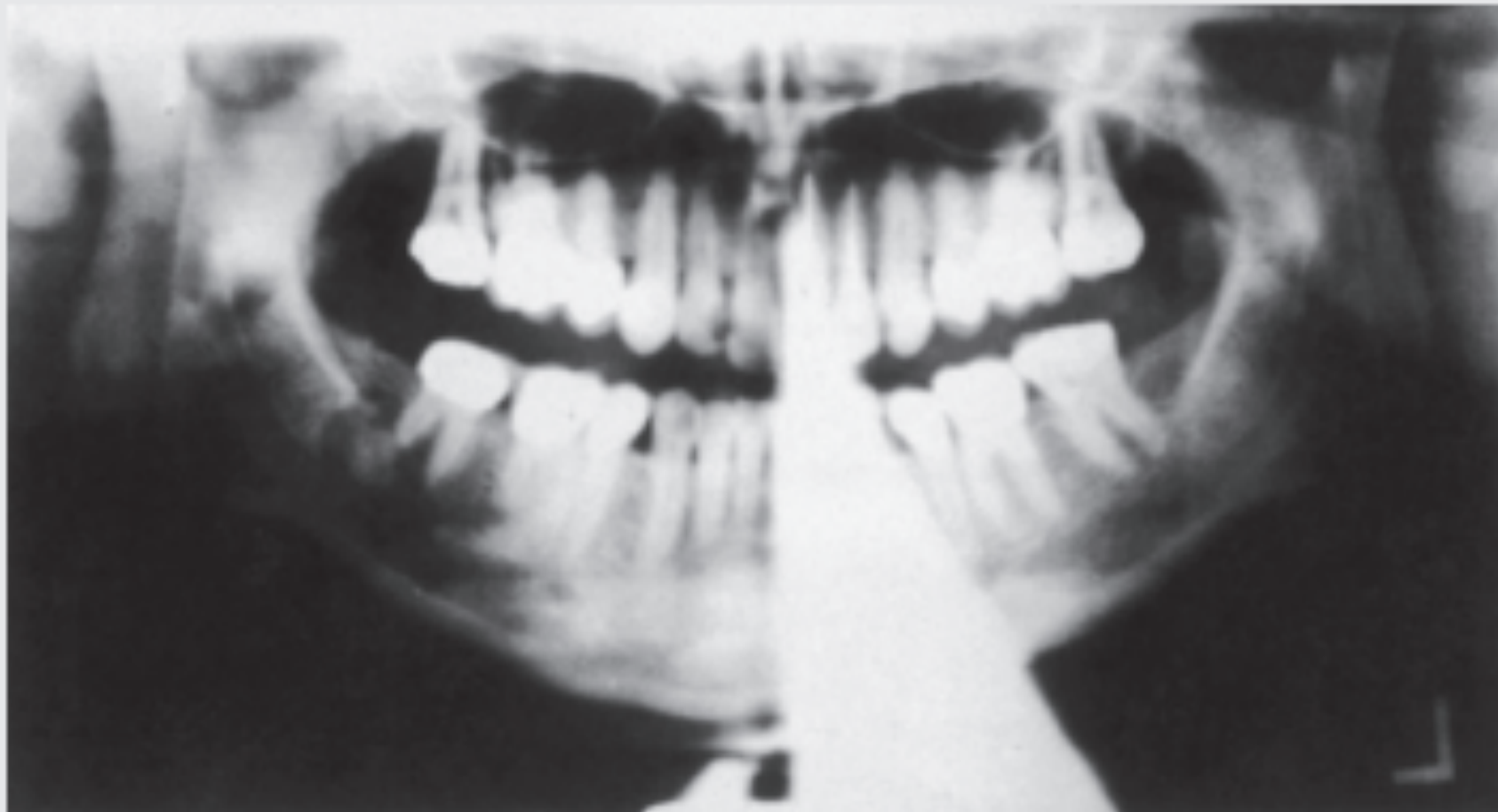
• **Figure 12.27** Patient positioned too far back. Upper and lower anterior teeth appear blurred and widened. **A**, Anterior teeth positioned behind the focal plane. **B**, Dental arch (*solid line*) posterior to focal plane (*broken line*). **C**, Radiograph.



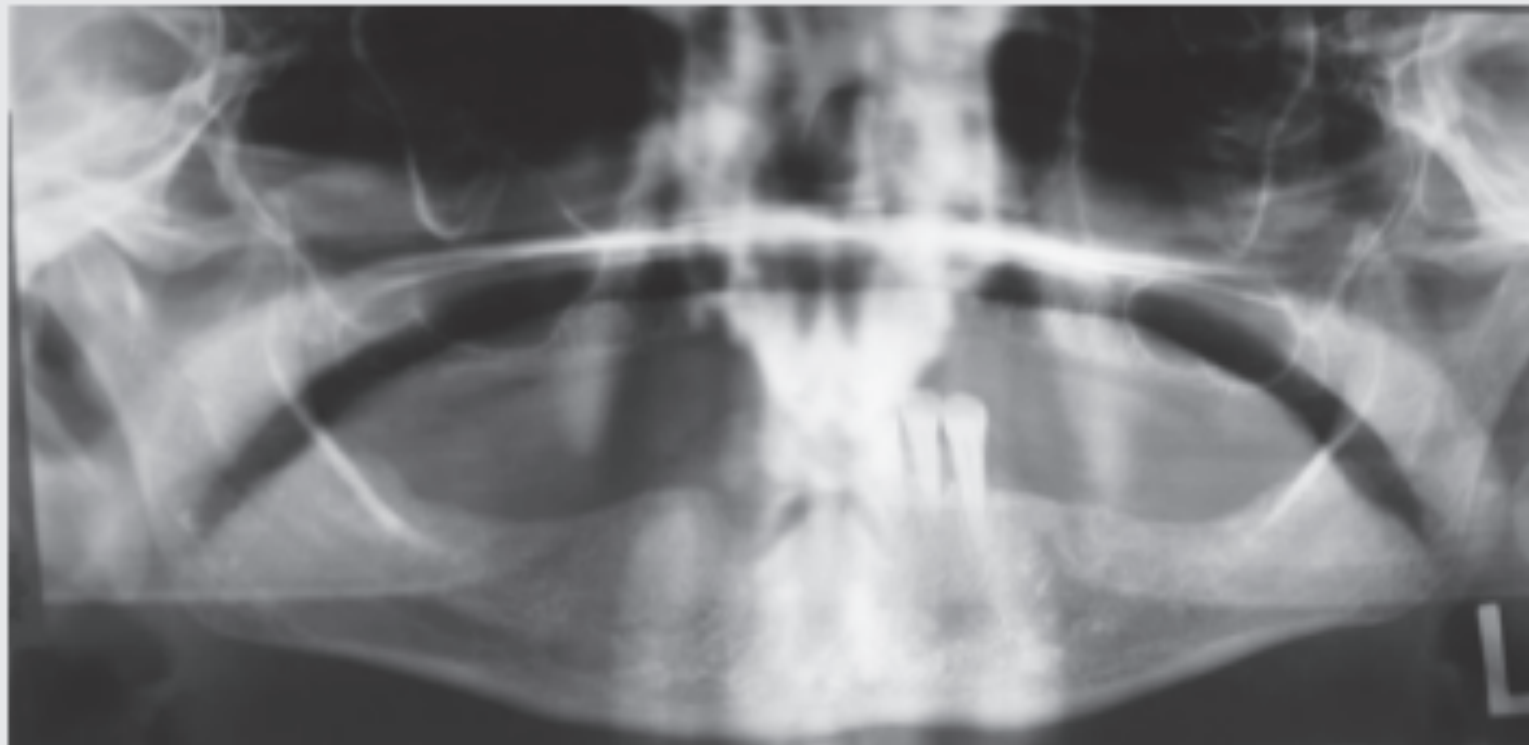
• **Figure 12.28** Patient's head tilted up. Hard palate (radiopaque band) image is superimposed on apices of upper teeth. **A**, Roots of maxillary incisors outside of focal plane. **B**, Mandibular arch (*unbroken line*) in focal trough. Maxillary arch (*broken line*) positioned in back of focal plane. **C**, Radiograph.



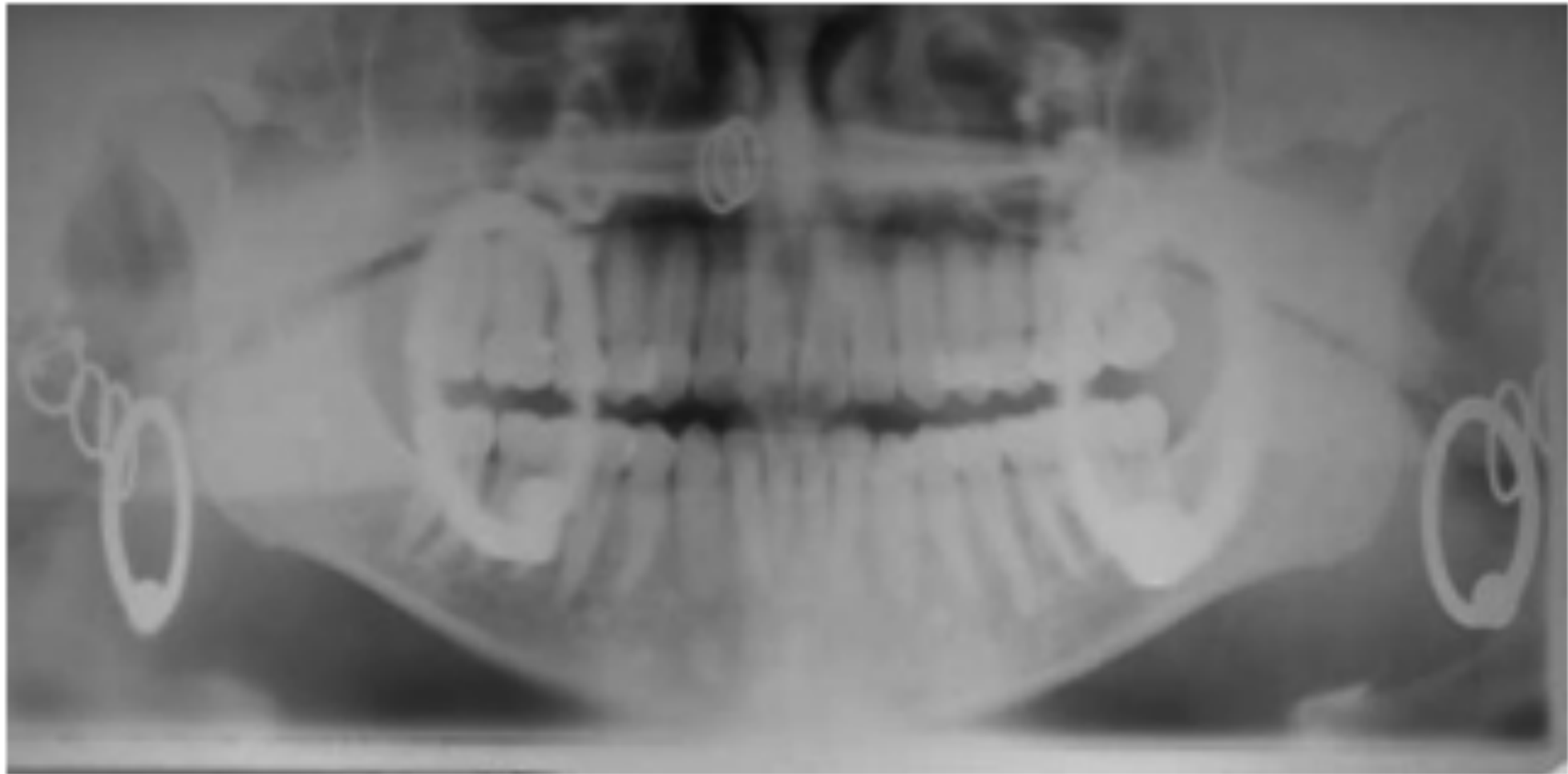
• **Figure 12.29** Patient's head is tilted down. Lower incisors are blurred and image of the hyoid bone is superimposed on the mandible. **A**, Roots of mandibular teeth outside of focal plane. **B**, Maxillary arch (unbroken line) positioned in focal plane. Mandibular arch (broken line) positioned in back of focal plane. **C**, Radiograph.



• **Figure 12.32** Patient slouching. Note superimposition of triangular radiopacity, representing the spinal



- **Figure 12.31** Tongue not against roof of the patient's mouth. Note the large radiolucent band superimposed over apices of the maxillary teeth.



- **Figure 12.37** Ghosting of various piercings on a panoramic radiograph.

Ghost image

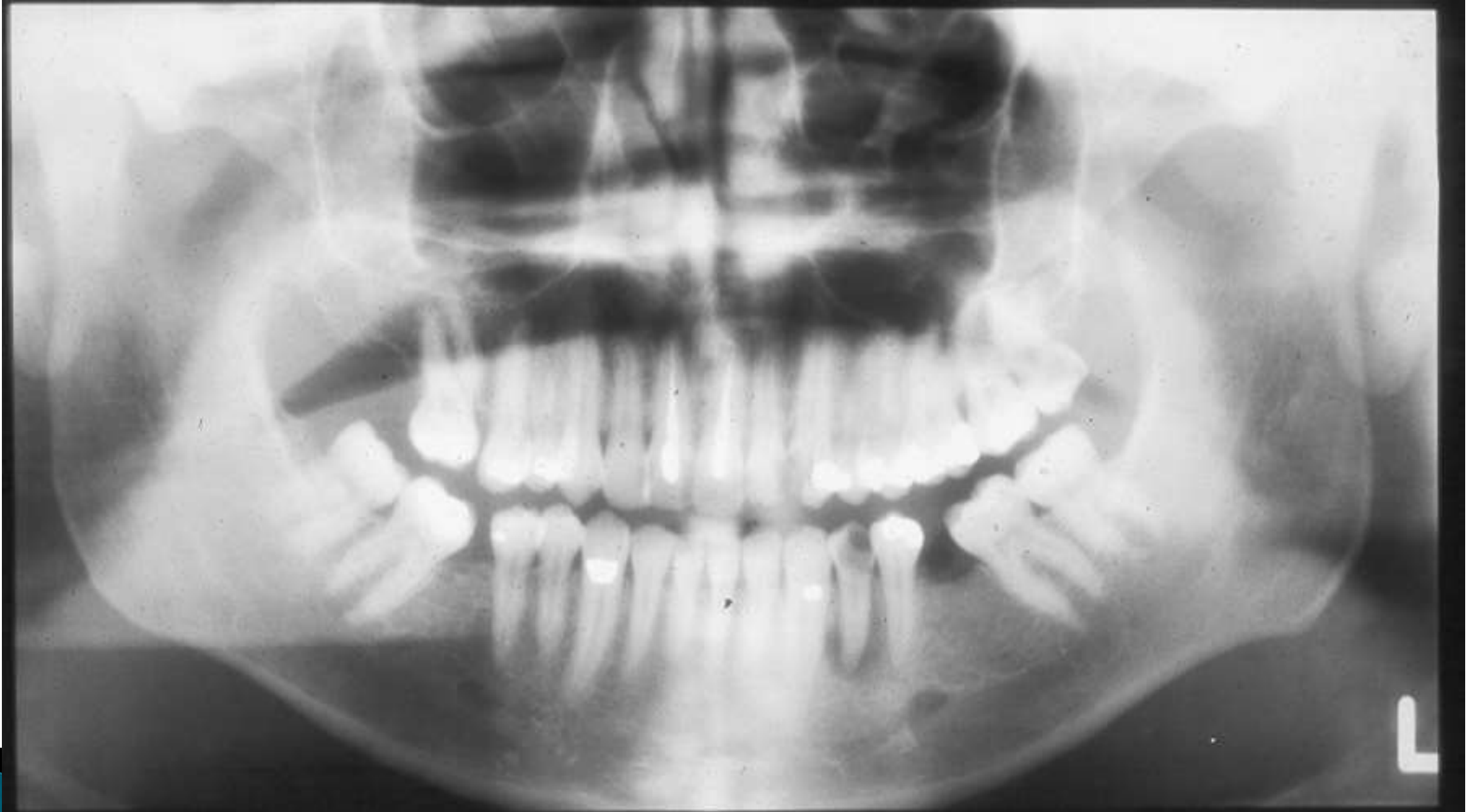


PATIENT POSITIONING ERRORS

POSITIONING OF THE LIPS AND TEETH

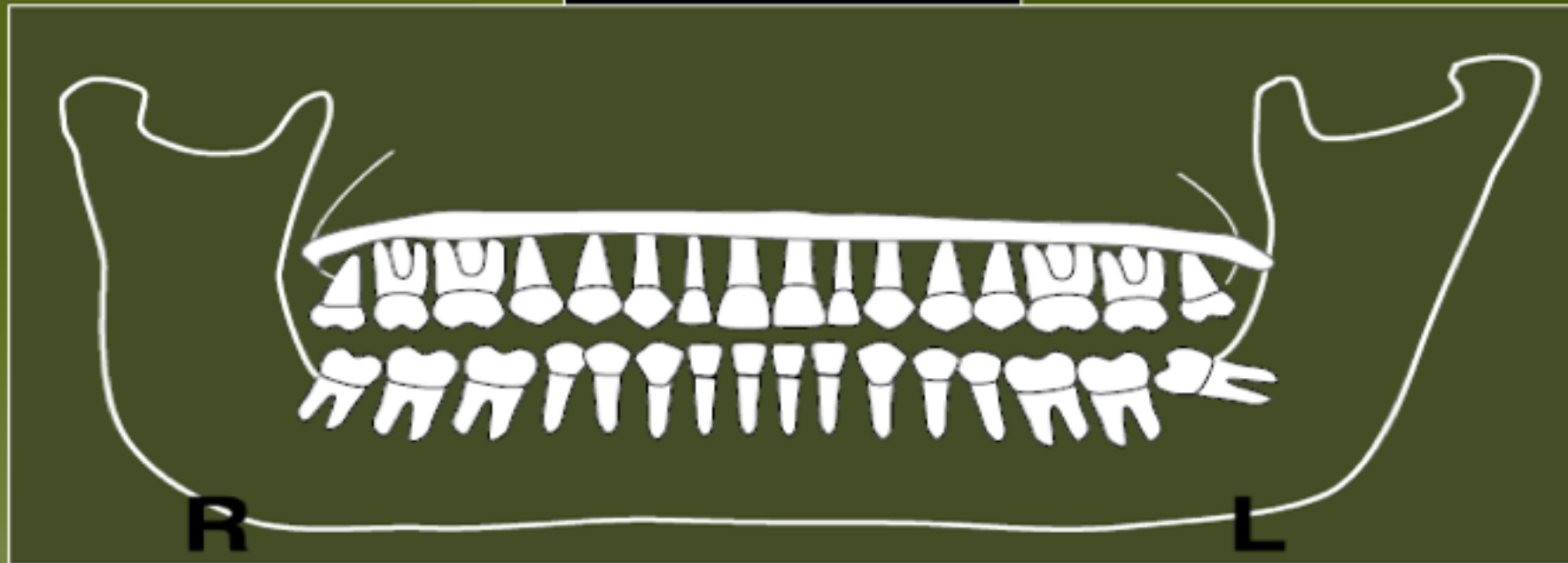
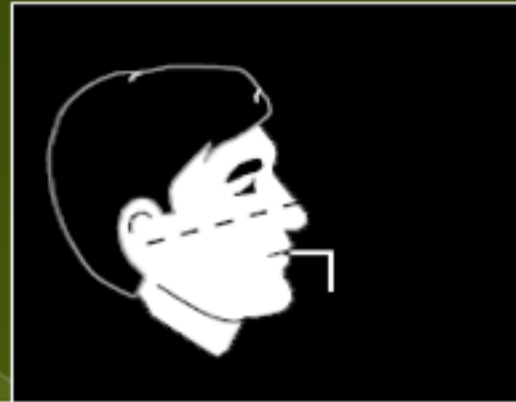
- ◎ If the lips are not closed on the bite block, a dark radiolucent shadow obscure the anterior teeth.
- ◎ If the tongue is not in contact with palate, a dark radiolucent shadow obscures the apices of the maxillary teeth.

Palatoglossal Air Space



Frankfort plane position

HEAD TIPPED UP



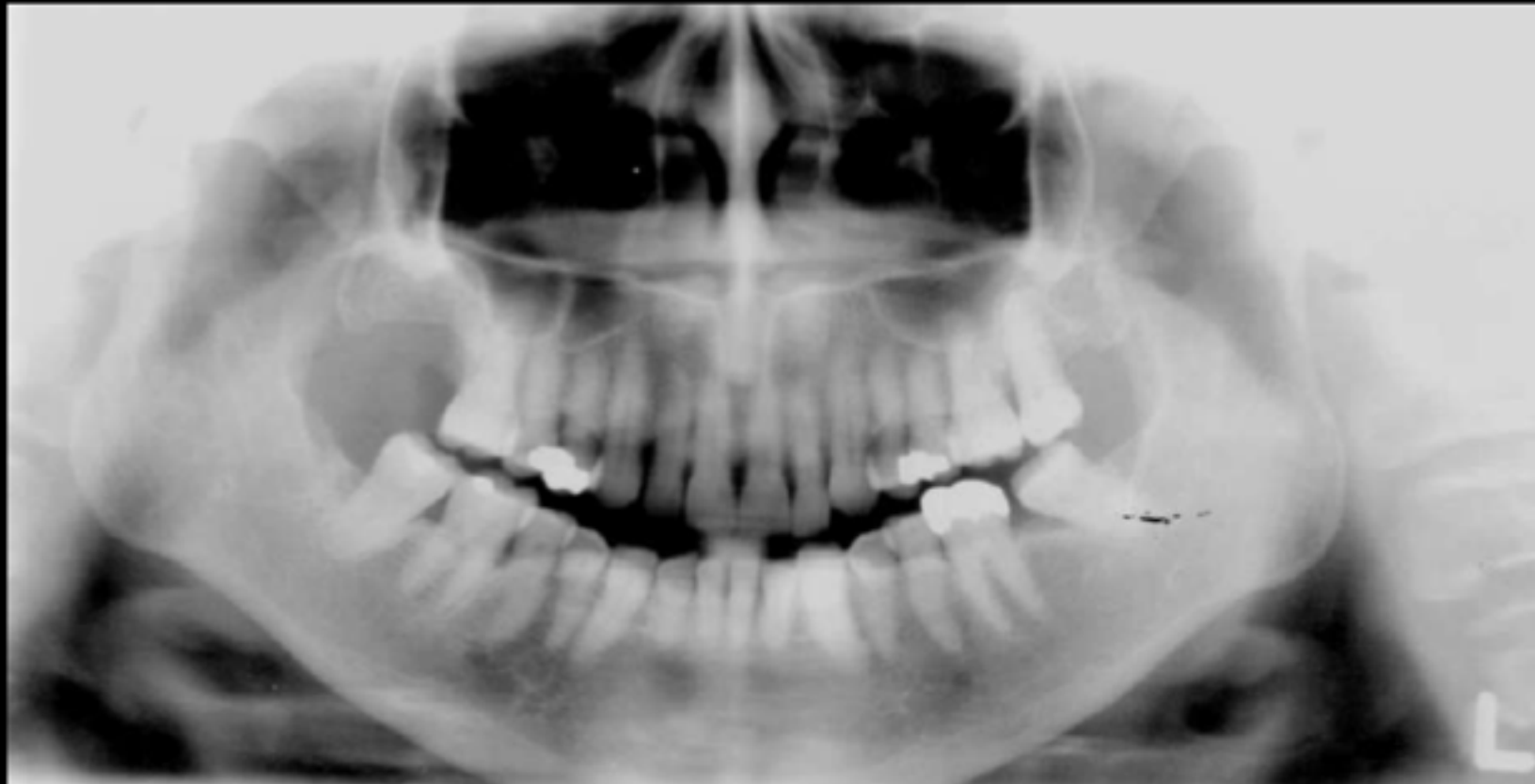
Squared-off mandible, palate superimposed over maxillary teeth

HEAD TIPPED DOWN



Mandibular incisors shortened, V-shaped mandible

HEAD TIPPED DOWN



TEETH TOO ANTERIOR



Anterior teeth narrower and blurred

Teeth too posterior



Anterior teeth wider and blurred

- ▶ it is important to know a good panoramic radiograph the mandible is "U" shaped, the condyles are positioned about an inch inside the edge of the film and $\frac{1}{3}$ of the way down from the top edge of the film. The occlusal plane exhibits a slight curve or "smile line", upwards. The roots of the maxillary & mandibular anterior teeth are readily visible.